

INTERACTIVE SPUTTERING SIMULATION

A browser-based reduced-order magnetron PVD simulator for undergraduate and postgraduate thin-film science education.

YAMAMURA-BOHDANSKY YIELD MODEL

REDUCED-ORDER ION TRANSPORT

THOMPSON ENERGY SAMPLING

THREE GROWTH MODES

REACTIVE SPUTTERING

CO-SPUTTERING • 3 GUNS

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FPS PHYSICS

0

INSTALLATION

DEPLOYMENT

Launch [index.html](#) in any modern browser — Chrome, Firefox, Edge, or Safari. No server, no installation, no dependencies beyond a browser. Runs fully offline. Tested on Windows, macOS, and Linux. Suitable for computer-lab sessions, self-directed study, and remote coursework delivery.

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PHYSICS ENGINE

CORE MODELS

YAMAMURA-BOHDANSKY SPUTTERING YIELD

$$Y(E) = Q \cdot S_n^*(E) \cdot (1 - (E_{th} / E)^{2/3})^{0.25}$$

Semi-empirical yield formula valid from threshold to keV regime. Q is a material-dependent constant encoding binding energy and mass ratio. $S_n^*(E)$ is the reduced nuclear stopping cross-section computed from the universal Lindhard-Scharff-Schiott (LSS) function. $E_{th} \approx U_s / \gamma$ where $\gamma = 4M_1M_2/(M_1+M_2)^2$ is the kinematic factor. Implemented for all 9 target materials with tabulated Q , U_s , E_{th} values.

COSINE-POWER EMISSION DISTRIBUTION

$$f(\theta) \propto \cos^n(\theta) \text{ where } n = 1 + 0.6 \cdot (E/500)^{0.4}$$

Angular distribution of sputtered atoms from the target surface under normal Ar^+ incidence. Pure cosine ($n = 1$) at low ion energies transitions to overcosine ($n > 1$) at higher energies, consistent with molecular dynamics and experimental observations. Controls the radial deposition footprint and uniformity profile on the substrate.

THOMPSON-SIGMUND ENERGY DISTRIBUTION

$$f(E_s) \propto E_s / (E_s + U_s)^3 \cdot (1 - \sqrt{(E_s + U_s) / E_{max}})$$

Energy spectrum of sputtered atoms at the target surface, derived from the linear collision cascade model. U_s is the surface binding energy. $E_{max} = \gamma E - U_s$ is the maximum transferable energy. Sampled stochastically each frame to assign initial kinetic energies to emitted atom packets.

MEAN FREE PATH - KINETIC THEORY

$$\lambda = kT / (\sqrt{2} \cdot \pi \cdot d_{Ar}^2 \cdot P)$$

Hard-sphere kinetic theory MFP for Ar at process temperature T and pressure P . $d_{Ar} = 3.7 \times 10^{-10}$ m. $k = 1.38 \times 10^{-23}$ J/K. Evaluated per frame from current pressure and gas mixture (Ar + O₂). The scattering ratio d/λ drives the transport regime classification: Ballistic ($d/\lambda < 0.5$), Moderate (0.5–2.0), High (> 2.0). Mixed Ar/O₂ atmospheres use a mole-fraction-weighted cross-section.

LORENTZ E×B ELECTRON DRIFT - MAGNETRON PRINCIPLE

$$v_{drift} = (E \times B) / B^2 \quad r_L = m_e \cdot v_{\perp} / (e \cdot B)$$

Secondary electrons emitted from the target are trapped in crossed electric and magnetic fields, executing cycloidal drift trajectories above the target surface. The closed drift path defines the racetrack erosion groove. Increased ionisation in the racetrack region raises local plasma density, sustaining discharge at lower pressures (1–10 mTorr) than a diode configuration.

PARTICLE TRANSPORT IMPLEMENTATION

ION DYNAMICS

Ar^+ ions spawn at the plasma-sheath boundary, accelerate toward the cathode under the sheath electric field with $E \times B$ lateral wobble. Impact is detected via a swept collision check (no tunnelling at high simulation speed). Ion energy at impact determines sputtering yield and ejected atom energy via Thompson sampling.

ATOM TRANSPORT & DEPOSITION

Sputtered atom packets travel upward with cosine-sampled angle and Thompson-sampled energy. Each frame computes scattering probability from λ . Successful substrate impacts are distributed across bins via a Gaussian kernel with racetrack shoulder rings. Sticking coefficient applied probabilistically. Particle arrays are ceiling-bounded to prevent frame rate collapse.

MATERIAL LIBRARY & PARAMETERS

TARGET MATERIAL DATABASE — YAMAMURA PARAMETERS

MATERIAL	SYMBOL	BINDING ENERGY U _S (EV)	THRESHOLD E _{TH} (EV)	YAMAMURA Q	YIELD @ 500 EV AR ⁺
Copper	Cu	3.49	17	0.356	~2.3
Aluminium	Al	3.36	13	0.241	~1.0
Titanium	Ti	4.85	20	0.179	~0.6
Silver	Ag	2.97	15	0.508	~3.1
Silicon	Si	4.63	25	0.071	~0.5
Tantalum	Ta	8.10	33	0.100	~0.3
Zinc	Zn	1.35	14	0.208	~3.8
Tin	Sn	3.12	16	0.290	~1.9
Tungsten	W	8.83	40	0.128	~0.4

CONFIGURABLE PROCESS PARAMETERS

POWER & DISCHARGE

Gun power per source	80 – 500 W
Power mode	DC / RF selectable
Active guns	1 – 3 independent
Discharge voltage (est.)	200 – 700 V
Ion energy at target	50 – 1000 eV

GEOMETRY

Target diameter	2 – 6 inches
Substrate diameter	2 – 8 inches
Target-substrate distance	5 – 25 cm
Substrate rotation	Continuous (speed-scaled)
Uniformity calculation	Live radial profile

GAS & PRESSURE

Working pressure	0.5 – 30 mTorr
Gas composition	Ar / Ar+O ₂ mixture
O ₂ fraction	0 – 50 %
MFP range (displayed)	~3 mm – 200 mm
Transport regime	Ballistic / Moderate / High

FILM GROWTH

Growth mode	FM / VW / SK / Auto
Lattice mismatch (SK)	0 – 15 %
Surface mobility	Low / Medium / High
Thickness target	1 – 5000 nm (auto-pause)
Simulation speed	0.5× – 5.0×

REACTIVE & CO-SPUTTERING CAPABILITIES

REACTIVE SPUTTERING (AR / O₂)

O₂ partial pressure modifies target poisoning level dynamically. Sputtering yield decreases as oxide forms on the target surface. Hysteresis indicator shows that return to metallic mode requires O₂ fraction below the onset value — faithfully reproducing the Berg hysteresis loop. RF mode shows reduced poisoning sensitivity relative to DC.

CO-SPUTTERING — 3 INDEPENDENT GUNS

Up to three guns operate simultaneously with independent material and power settings. Film composition is tracked proportionally using per-source yield values. Alloy fraction readout updates in real time. Demonstrates why equal power does not equal equal composition due to yield differences — a key industrial concept for ITO, Cu-Al, and multi-component alloy deposition.

TUTORIAL CURRICULUM & LAB MODULES

14-CHAPTER TUTORIAL — SEQUENTIAL LEARNING PATHWAY

01 Why Thin Films? FOUNDATION	02 Vacuum Science PRESSURE & PUMPS	03 Why Argon? GAS SELECTION	04 What is Sputtering? INTRODUCTION
05 Chamber Components HARDWARE	06 Plasma Formation DC & RF	07 Magnetron Design E×B DRIFT	08 Sputtering Yield BOHDANSKY
09 Mean Free Path GAS TRANSPORT	10 Film Uniformity DEPOSITION	11 Reactive Sputtering TiN & HYSTERESIS	12 Co-Sputtering MULTI-GUN
13 Film Growth Modes FM · VW · SK	14 Film Characterisation METROLOGY		

Each chapter includes: physics explanation · animated SVG diagrams · governing equations · real-world reference with peer-reviewed citation

5 STRUCTURED VIRTUAL LAB MODULES

LAB 01 Pressure vs. Transport Regime Vary working pressure from 1.5 to 18 mTorr and observe the transition from ballistic to high-scattering transport. Measure deposition rate, scattering ratio d/λ , and radial uniformity at each pressure point. Export time-series CSV and identify the crossover pressure where $d/\lambda = 1$. VARIABLE: PRESSURE ~30 MIN CSV EXPORT REQUIRED
LAB 02 Sputtering Yield vs. Power & Material Compare yield response across Cu, Al, Zn, W, and Ta as gun power increases from 100 W to 500 W. Quantify the role of binding energy and mass ratio in determining yield. Identify which material shows the steepest power-yield gradient and explain using the Yamamura formula. VARIABLE: POWER + MATERIAL ~35 MIN MULTI-RUN COMPARISON
LAB 03 Geometry Optimisation for Uniformity Systematically vary target diameter, substrate diameter, and target-substrate distance to minimise non-uniformity percentage. Record the radial profile CSV at each condition. Identify the geometry combination that achieves the best uniformity and rationalise using the cosine emission footprint. VARIABLE: GEOMETRY ~40 MIN RADIAL PROFILE ANALYSIS
LAB 04 Reactive Sputtering Hysteresis Increase O ₂ fraction incrementally and record yield and deposition rate. Identify the poisoning onset point. Then reduce O ₂ and record the return path. Map the full hysteresis loop and compare DC vs. RF mode behaviour. Discuss implications for process control in TiN and ZnO deposition. VARIABLE: O ₂ FRACTION ~35 MIN HYSTERESIS MAPPING

LAB 05 Co-Sputtering Composition Control

FILM GROWTH MODES & FEATURES

THREE IMPLEMENTED GROWTH MODES

FRANK-VAN DER MERWE

Layer-by-layer growth. High surface mobility. Each monolayer completes before the next nucleates. Produces atomically flat, epitaxial films. Simulator shows sequential layer filling with uniform substrate coverage.

Condition: $E_{\text{adsorption}} > E_{\text{adhesion}}$
· Lattice mismatch $\sim 0\%$

VOLMER-WEBER

Island growth. Low surface mobility. Atoms prefer bonding to each other over the substrate. Random 3D islands nucleate and grow before coalescence. Simulator renders discrete nucleation sites that merge into a continuous film.

Condition: $E_{\text{adhesion}} > E_{\text{adsorption}}$
· High surface energy mismatch

STRANSKI-KRASTANOV

Layer then islands. Flat wetting layers form first, then accumulated strain from lattice mismatch drives 3D island nucleation. Simulator transitions mid-run at the critical thickness controlled by the lattice mismatch slider.

Condition: Small lattice mismatch initially · Strain energy accumulates

KEY PEDAGOGICAL FEATURES

AI RECIPE ADVISOR

An integrated advisory engine evaluates the current parameter set and flags configurations unlikely to produce a physically meaningful result. It suggests corrected recipes with predicted transport regime, uniformity percentage, and growth mode. Students compare their own prediction to the advisor output — a structured pre-lab exercise in process reasoning.

THICKNESS TARGET WITH AUTO-PAUSE

Students set a target film thickness in nm. A real-time progress bar tracks deposition toward the target. The simulator auto-pauses when the target is reached and displays an achievement summary: final thickness, elapsed time, uniformity percentage, and average deposition rate. Resume continues deposition beyond the target.

CSV DATA EXPORT

Three export formats: time-series log (rate, yield, MFP, uniformity per second), radial thickness profile (binned substrate cross-section), and process summary (all conditions at end of run). Enables quantitative post-session data analysis, graphing assignments, and direct comparison across experimental conditions.

INSTRUCTOR GUIDE DOWNLOAD

A one-click plain-text instructor guide generates dynamically from the lab module database. It includes learning objectives, expected trends, procedure steps, reflection questions with model answers, and a 10-mark rubric per module. Generated at runtime — always matches the current version of the lab modules.

REAL-WORLD REFERENCE CALLOUTS

Each of the 14 tutorial chapters contains a highlighted callout box linking the simulated concept to a published experimental result or industry practice, with a peer-reviewed citation. Examples include the IBM Cu interconnect paper (Edelstein 1999), the Berg reactive sputtering hysteresis paper (1988), and the Thornton zone model (1974).

REVEAL-ON-DEMAND EXPECTED RESULTS

Each lab module contains a hidden "What you should observe" panel that students unlock after completing their own observations. The panel provides the expected trend, the quantitative crossover value (e.g. $d/\lambda = 1$ at ~ 6 mTorr), and the physical mechanism in one paragraph. Prevents premature answer exposure while providing immediate feedback after the experiment.

PROCESS DIAGNOSTICS PANEL

Live status bar displays: discharge voltage (V) · ion current density (mA/cm^2) · ion energy (eV) · sputtering yield (atoms/ion) · illustrative yield (transport-corrected) · mean free path λ (cm) · scattering ratio d/λ · transport regime label · deposition rate (arb. units/s) · film thickness (nm) · non-uniformity (%) · growth mode · sticking coefficient · total process time. All values update at the chart refresh rate (approximately every 1 second of simulation time).

SCIENTIFIC REFERENCES

FOUNDATIONAL PHYSICS — SPUTTERING YIELD THEORY

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MAGNETRON SPUTTERING & PLASMA PHYSICS

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FILM GROWTH, TRANSPORT & CHARACTERISATION

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- [12] Depla, D. & Mahieu, S. (eds.) — *Reactive Sputter Deposition*. Springer Series in Materials Science, Vol. 109, 2008. Reactive gas modelling, target poisoning mechanisms, and process window control.

HOW TO CITE THIS SOFTWARE

CITATION

Gururajan, B. (2025). Interactive Sputtering Simulation (Version 1) [Educational Software]. Department of Electrical Engineering, Yuan Ze University, Taiwan.

MODEL LIMITATIONS

This simulator is built for learning and intuition, not fab-grade process fitting. The discharge model is a parametric fit — not a full PIC or fluid plasma solver. Atom trajectories use a reduced-order cosine emission and Thompson energy sampler, not molecular dynamics or SRIM/TRIM. Deposition rate and film thickness numbers are scaled for classroom